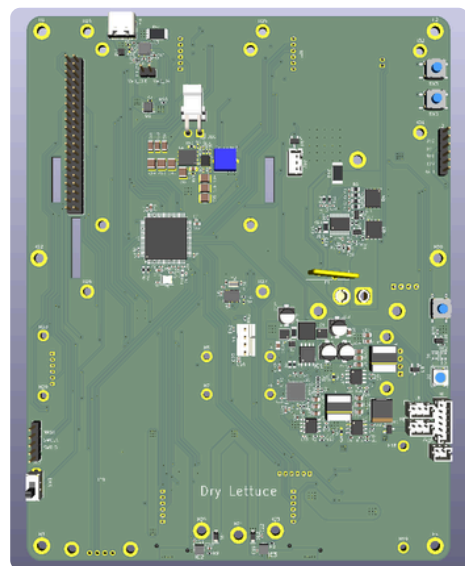


Hardware

We designed a PCB with some components being soldered in-house. This allowed for space saving and more reliable power management systems. It also reduced hardware complexities once it was working. The main motivation for this was gaining more experience with circuit design and layout.



VL53L1X Time of Flight Sensor

2x: One primary sensor at the front, and one on the claw for verifying victim capture

HC-SR04 Ultrasonic Sensor

Mounted on the side and used for navigating around obstacles

STM32F407 Microcontroller

Secondary processing unit for controlling motors, servos and illumination LEDs



SG90 Servos

3x: claw, claw lift and ball storage release

Pi Camera Module 3 Wide

2x for line follow and one front facing for rescue to detect evacuation points and victims

Pololu 20mm Metal Gear Motors

4x each with magnetic encoders for accurate speed and distance

OLED Display

Shows Pi & STM32 temperatures, ROS output for debugging, a status code, and turn angle

80mm Pololu Wheels

Larger wheels compared to previous designs to get over speed bumps better

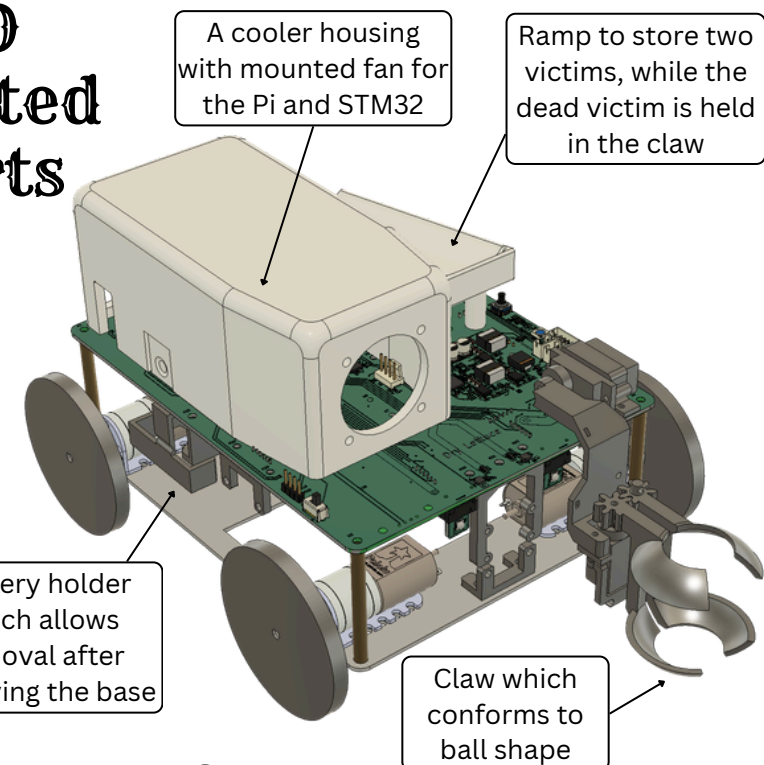
Raspberry Pi AI Hat+ 2

Offloads processing of victim recognition machine learning

Raspberry Pi 5

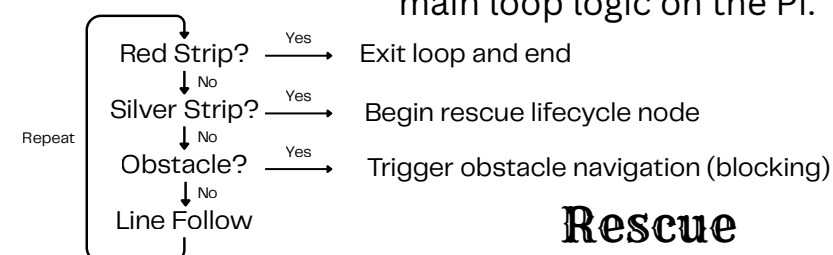
Main processing unit with 8GB of RAM

3D Printed Parts

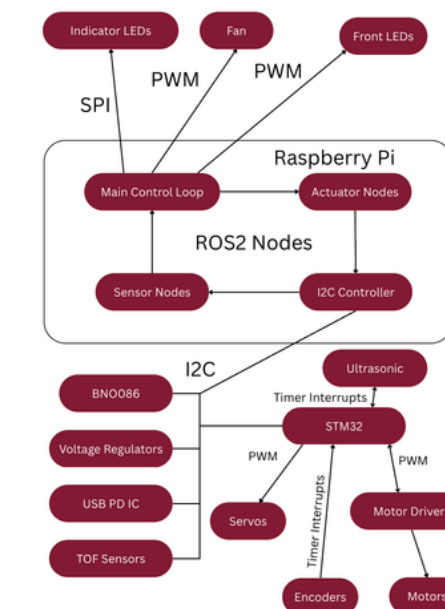


Software

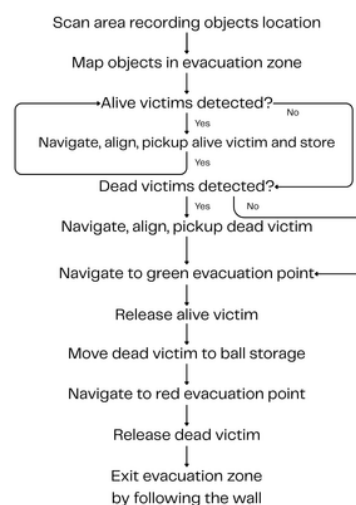
Below is the high level main loop logic on the Pi.



Below is a diagram of the ROS2 node connections

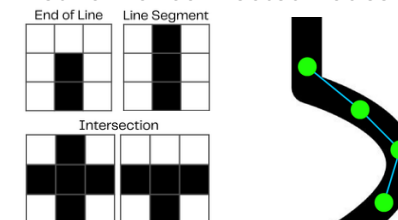


Rescue



Line Follow

The robot's algorithm constructs a graph-based map rather than simply tracking the largest contour. It compares intersections with the line position and links these into a network of connected nodes.



Past Achievements

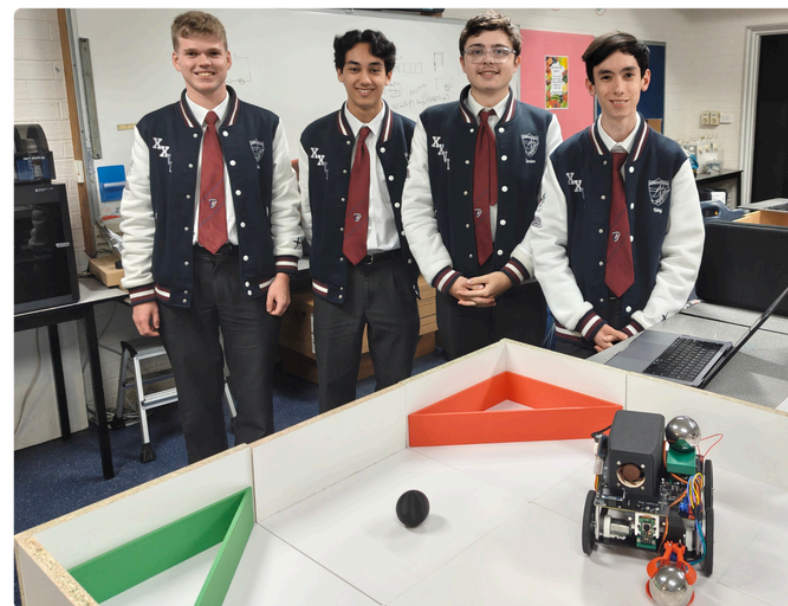
Our achievements from competing at RoboCup Junior Australia:

- 1st in the NSW state competition in 2022, 2023 and 2025, and 2nd in 2024
- 1st in the Australian National competition in 2025, 3rd in 2022 and 2023, and 2nd in 2024



Team

We have been a team since 2022, with Toby, Landon and Bain doing robotics since 2017, and William joining our school's robotics program in 2021.



William D'Olier
Left most

PCB Design

- Routing
- STM32
- Motor Drivers

Line Follow

- Line follow and green turn program

Motor Driver

- Programming for movement for motor drivers
- Servo control code

Documentation

- TDP

Toby McFall

Right most

PCB Design

- LED driver
- Fan connection
- Sensor connection
- Routing

CAD

- Robot structure
- Sensor mounts
- Cooler housing
- Ball ramp

Rescue

- Victim detection with machine learning model

Documentation

- BOM

Landon Aitken-Ling

Second from the right

Project Management

- Time plan
- Parts plan

Documentation

- Poster
- TDP

Software

- Bug testing/iterative coding of rescue zone
- Obstacle detection and readjusting to line

Bain Changson

Second from the left

PCB

- Power management
- Routing

Rescue

- Main rescue logic
- Mapping and navigating rescue zone

- Claw & lift mechanism
- Ball ramp
- Sensor mounts
- Battery holder